

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

```
In [2]: df = pd.read_csv("country_wise_latest.csv")

print(df.head())
print(df.info())
```

Out:

	Country/Region	Confirmed	Deaths	Recovered	Active	New cases	New deaths	\
0	Afghanistan	36263	1269	25198	9796	106	10	
1	Albania	4880	144	2745	1991	117	6	
2	Algeria	27973	1163	18837	7973	616	8	
3	Andorra	907	52	803	52	10	0	
4	Angola	950	41	242	667	18	1	

	New recovered	Deaths / 100 Cases	Recovered / 100 Cases	\
0	18	3.50	69.49	
1	63	2.95	56.25	
2	749	4.16	67.34	
3	0	5.73	88.53	
4	0	4.32	25.47	

	Deaths / 100 Recovered	Confirmed last week	1 week change	\
0	5.04	35526	737	
1	5.25	4171	709	
2	6.17	23691	4282	
3	6.48	884	23	
4	16.94	749	201	

	1 week % increase	WHO Region
0	2.07	Eastern Mediterranean
1	17.00	Europe
2	18.07	Africa
3	2.60	Europe
4	26.84	Africa

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 187 entries, 0 to 186

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	Country/Region	187 non-null	object
1	Confirmed	187 non-null	int64
2	Deaths	187 non-null	int64
3	Recovered	187 non-null	int64
4	Active	187 non-null	int64
5	New cases	187 non-null	int64
6	New deaths	187 non-null	int64
7	New recovered	187 non-null	int64
8	Deaths / 100 Cases	187 non-null	float64
9	Recovered / 100 Cases	187 non-null	float64
10	Deaths / 100 Recovered	187 non-null	float64
11	Confirmed last week	187 non-null	int64
12	1 week change	187 non-null	int64
13	1 week % increase	187 non-null	float64
14	WHO Region	187 non-null	object

dtypes: float64(4), int64(9), object(2)

memory usage: 22.0+ KB

None

In [3]: `print(df.isnull().sum())`

Out:

Country/Region	0
Confirmed	0
Deaths	0
Recovered	0
Active	0
New cases	0
New deaths	0
New recovered	0
Deaths / 100 Cases	0
Recovered / 100 Cases	0
Deaths / 100 Recovered	0
Confirmed last week	0
1 week change	0
1 week % increase	0
WHO Region	0

dtype: int64

In [4]: `print(df.duplicated().sum())`

Out:

0

```
In [5]: df = df.drop_duplicates()
```

```
In [6]: print(df.columns)
```

Out:

```
Index(['Country/Region', 'Confirmed', 'Deaths', 'Recovered', 'Active',  
      'New cases', 'New deaths', 'New recovered', 'Deaths / 100 Cases',  
      'Recovered / 100 Cases', 'Deaths / 100 Recovered',  
      'Confirmed last week', '1 week change', '1 week % increase',  
      'WHO Region'],  
      dtype='object')
```

```
In [7]: print(df.shape)  
        print(df.head())
```

Out:

(187, 15)

	Country/Region	Confirmed	Deaths	Recovered	Active	New cases	New deaths	\
0	Afghanistan	36263	1269	25198	9796	106	10	
1	Albania	4880	144	2745	1991	117	6	
2	Algeria	27973	1163	18837	7973	616	8	
3	Andorra	907	52	803	52	10	0	
4	Angola	950	41	242	667	18	1	

	New recovered	Deaths / 100 Cases	Recovered / 100 Cases	\
0	18	3.50	69.49	
1	63	2.95	56.25	
2	749	4.16	67.34	
3	0	5.73	88.53	
4	0	4.32	25.47	

	Deaths / 100 Recovered	Confirmed last week	1 week change	\
0	5.04	35526	737	
1	5.25	4171	709	
2	6.17	23691	4282	

3	6.48	884	23
4	16.94	749	201

	1 week % increase	WHO Region
0	2.07	Eastern Mediterranean
1	17.00	Europe
2	18.07	Africa
3	2.60	Europe
4	26.84	Africa

```
In [8]: top10 = df.sort_values('Confirmed', ascending=False).head(10)

print(top10[['Country/Region', 'Confirmed']])
```

Out:

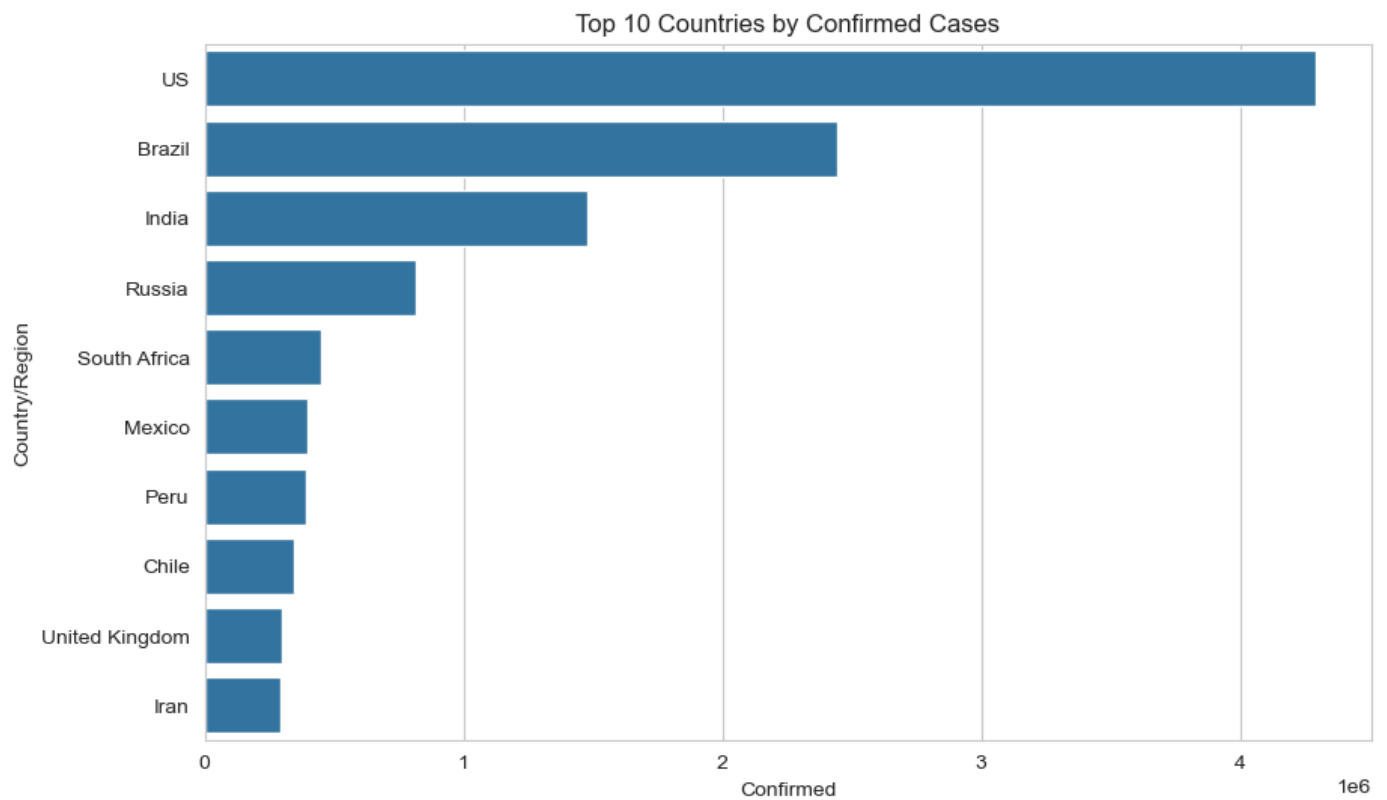
	Country/Region	Confirmed
173	US	4290259
23	Brazil	2442375
79	India	1480073
138	Russia	816680
154	South Africa	452529
111	Mexico	395489
132	Peru	389717
35	Chile	347923
177	United Kingdom	301708
81	Iran	293606

```
In [9]: import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))
sns.barplot(data=top10,
            x='Confirmed',
            y='Country/Region')

plt.title('Top 10 Countries by Confirmed Cases')
plt.show()
```

Out :

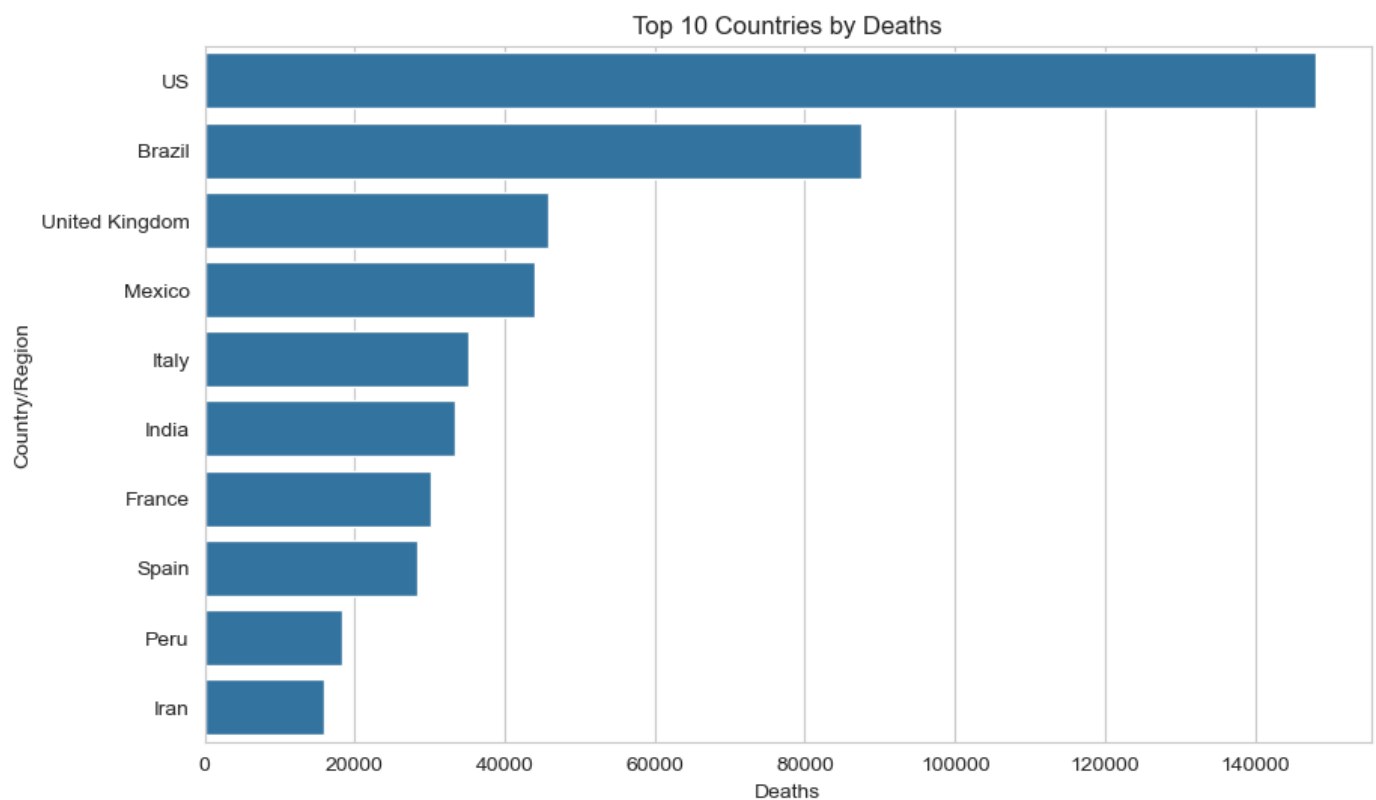


```
In [10]: top10_deaths = df.sort_values('Deaths', ascending=False).head(10)
```

```
plt.figure(figsize=(10,6))
sns.barplot(data=top10_deaths,
            x='Deaths',
            y='Country/Region')

plt.title('Top 10 Countries by Deaths')
plt.show()
```

Out:



```
In [11]: df['Recovery Rate'] = (
            df['Recovered'] / df['Confirmed']
        ) * 100

top_recovery = df.sort_values(
    'Recovery Rate',
    ascending=False
).head(10)

print(top_recovery[
    ['Country/Region', 'Recovery Rate']
])
```

Out:

	Country/Region	Recovery Rate
75	Holy See	100.000000
69	Grenada	100.000000
49	Dominica	100.000000
48	Djibouti	98.379126
78	Iceland	98.327940
24	Brunei	97.872340
121	New Zealand	97.238279
136	Qatar	97.017254

```
105      Malaysia      96.597035
110      Mauritius      96.511628
```

```
In [12]: print(df.head())
```

Out:

	Country/Region	Confirmed	Deaths	Recovered	Active	New cases	New deaths	\
0	Afghanistan	36263	1269	25198	9796	106	10	
1	Albania	4880	144	2745	1991	117	6	
2	Algeria	27973	1163	18837	7973	616	8	
3	Andorra	907	52	803	52	10	0	
4	Angola	950	41	242	667	18	1	

	New recovered	Deaths / 100 Cases	Recovered / 100 Cases	\
0	18	3.50	69.49	
1	63	2.95	56.25	
2	749	4.16	67.34	
3	0	5.73	88.53	
4	0	4.32	25.47	

	Deaths / 100 Recovered	Confirmed last week	1 week change	\
0	5.04	35526	737	
1	5.25	4171	709	
2	6.17	23691	4282	
3	6.48	884	23	
4	16.94	749	201	

	1 week % increase	WHO Region	Recovery Rate
0	2.07	Eastern Mediterranean	69.486805
1	17.00	Europe	56.250000
2	18.07	Africa	67.339935
3	2.60	Europe	88.533627
4	26.84	Africa	25.473684

```
In [13]: top_active = df.sort_values(
            'Active',
            ascending=False
        ).head(10)

print(top_active[
```

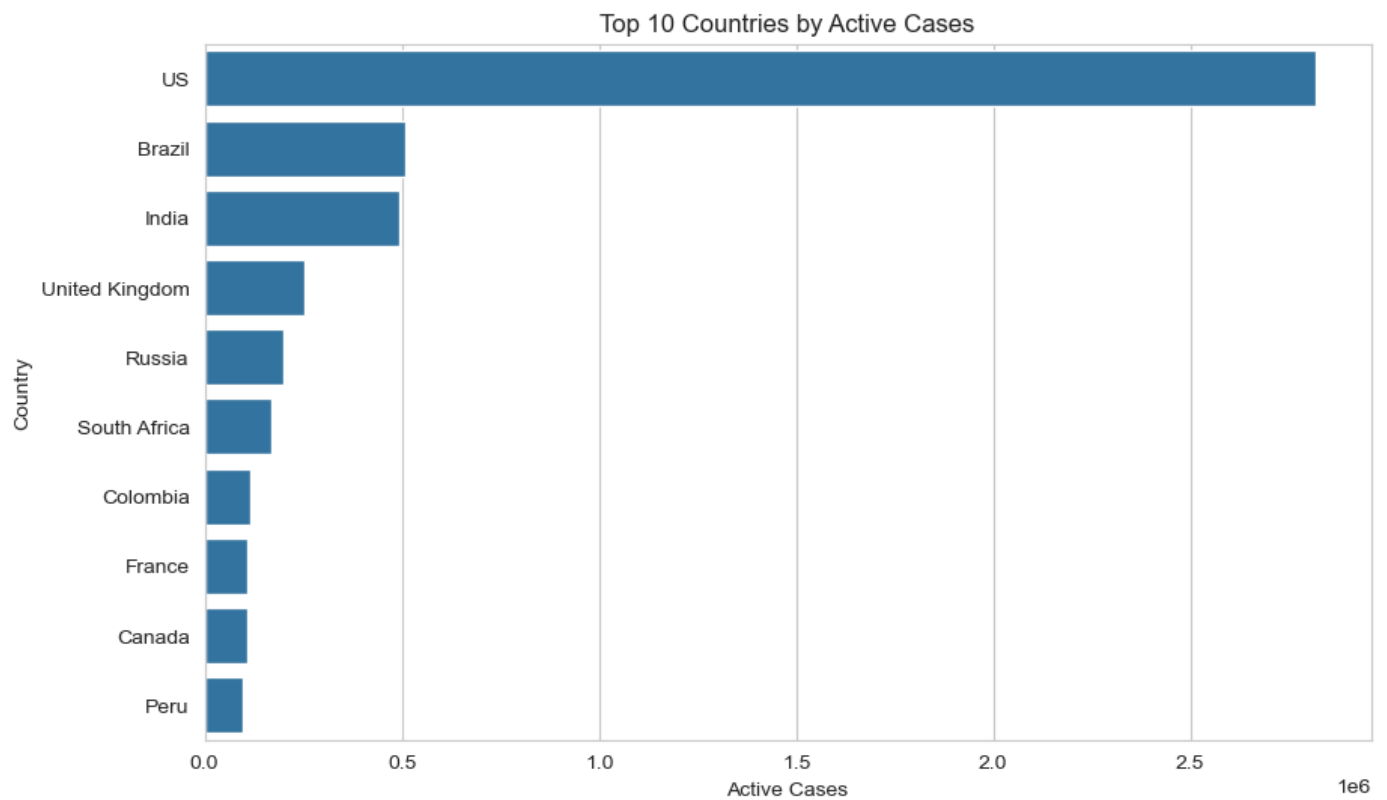
```
['Country/Region', 'Active']  
])
```

Out:

	Country/Region	Active
173	US	2816444
23	Brazil	508116
79	India	495499
177	United Kingdom	254427
138	Russia	201097
154	South Africa	170537
37	Colombia	117163
61	France	108928
32	Canada	107514
132	Peru	98752

```
In [14]: plt.figure(figsize=(10,6))  
  
sns.barplot(  
    data=top_active,  
    x='Active',  
    y='Country/Region'  
)  
  
plt.title('Top 10 Countries by Active Cases')  
plt.xlabel('Active Cases')  
plt.ylabel('Country')  
plt.show()
```


Out:



```
In [15]: region_cases = df.groupby(
            'WHO Region'
        )['Confirmed'].sum()

        print(region_cases)
```

Out:

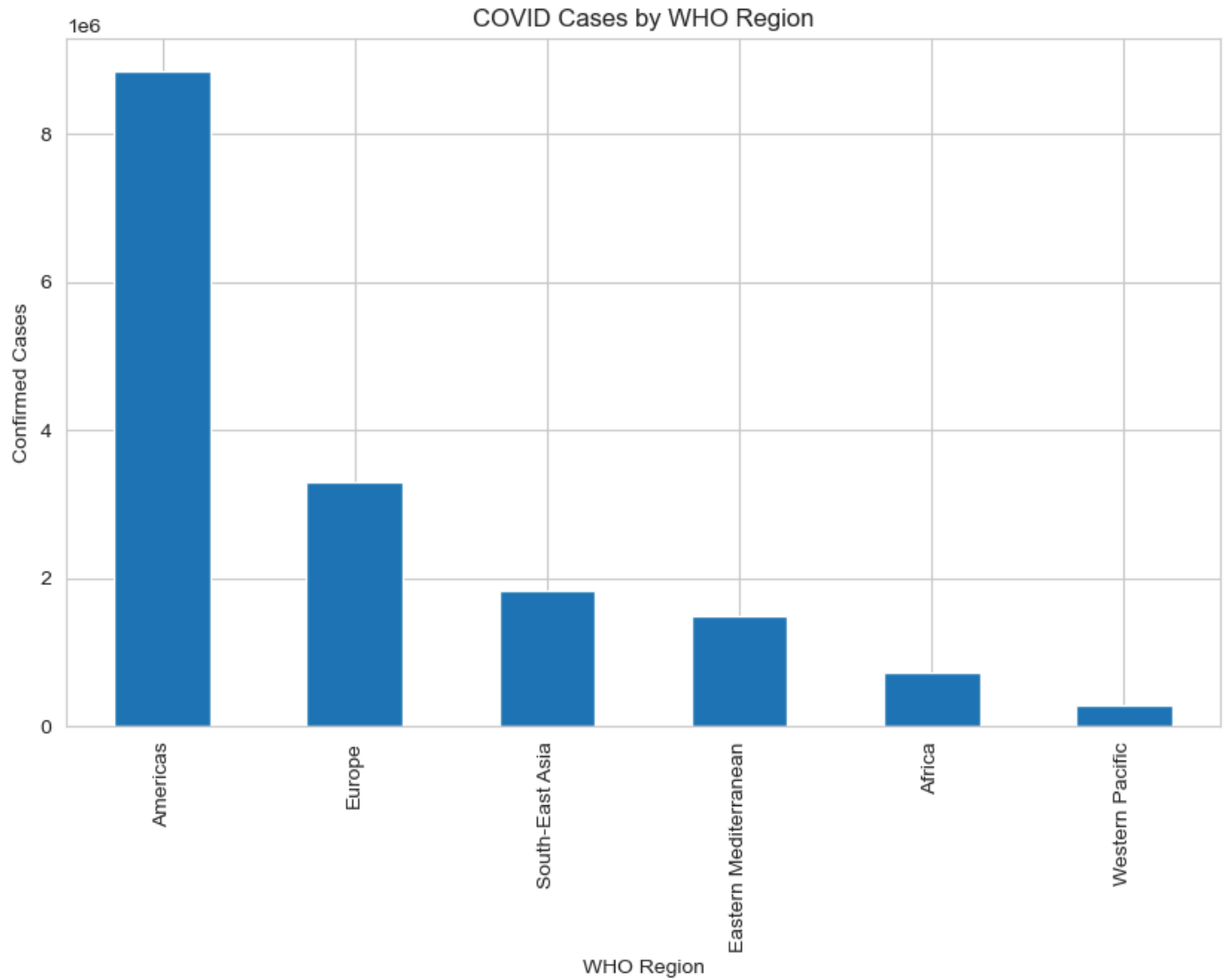
```
WHO Region
Africa          723207
Americas        8839286
Eastern Mediterranean  1490744
Europe          3299523
South-East Asia  1835297
Western Pacific  292428
Name: Confirmed, dtype: int64
```

```
In [16]: plt.figure(figsize=(10,6))

        region_cases.sort_values(
            ascending=False
        ).plot(kind='bar')
```

```
plt.title('COVID Cases by WHO Region')  
plt.xlabel('WHO Region')  
plt.ylabel('Confirmed Cases')  
plt.show()
```

Out:



```
In [17]: correlation = df[  
        ['Confirmed',  
         'Deaths',  
         'Recovered',  
         'Active']  
        ].corr()  
  
print(correlation)
```

Out:

	Confirmed	Deaths	Recovered	Active
Confirmed	1.000000	0.934698	0.906377	0.927018
Deaths	0.934698	1.000000	0.832098	0.871586
Recovered	0.906377	0.832098	1.000000	0.682103
Active	0.927018	0.871586	0.682103	1.000000

```
In [18]: plt.figure(figsize=(8,6))

sns.heatmap(
    correlation,
    annot=True,
    cmap='Blues'
)

plt.title('Correlation Heatmap')
plt.show()
```

Out :

